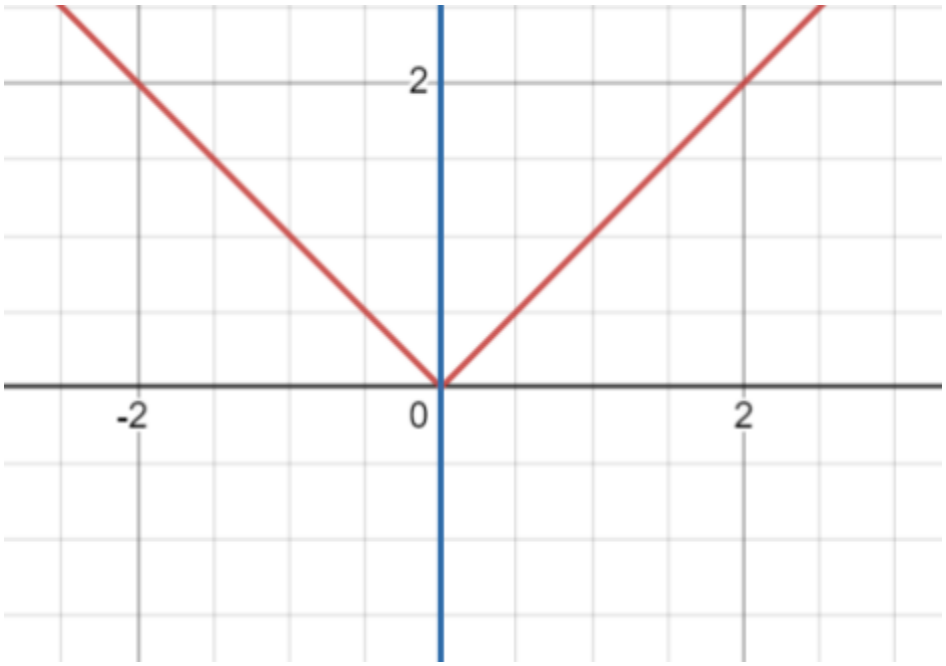
ECE330 Homework 6

a) Draw graphically and explain in words four qualitatively different cases where the 1D Newton--Raphson Root Finding Method will fail. (1.2 points each, 4.8 total)

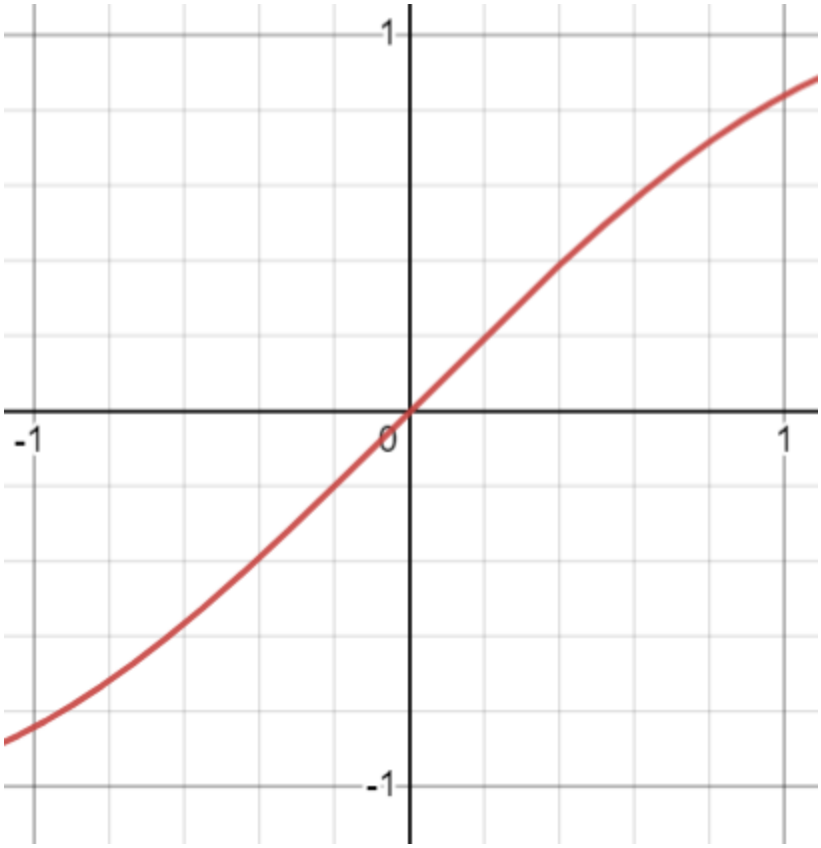
The Newton-Raphson method is a numerical root-finding algorithm that uses an iterative process to approximate the roots of a given function. Here are four qualitatively different cases where the Newton-Raphson method may fail:

1. The function has a vertical tangent at the root: In this case, the derivative of the function is undefined at the root, which makes it impossible for the Newton-Raphson method to converge. For example, the function f(x) = |x| has a vertical tangent at x = 0, which means that the Newton-Raphson method cannot find the root at x = 0.



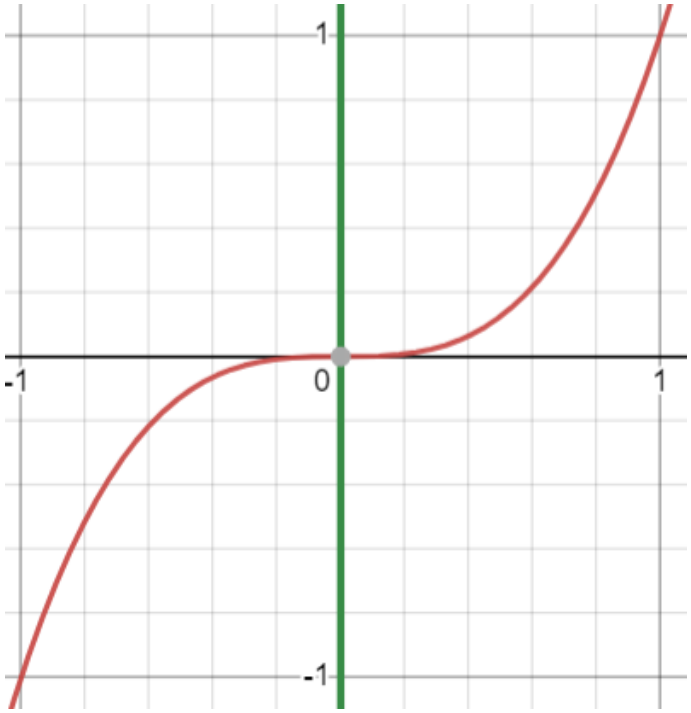
f(x) = |x| at x = 0

2. The function oscillates near the root: In this case, the Newton-Raphson method may converge to a root, but it may also oscillate back and forth between two or more roots. For example, the function f(x) = sin(x) has infinitely many roots, and the Newton-Raphson method may converge to any one of them depending on the initial guess.



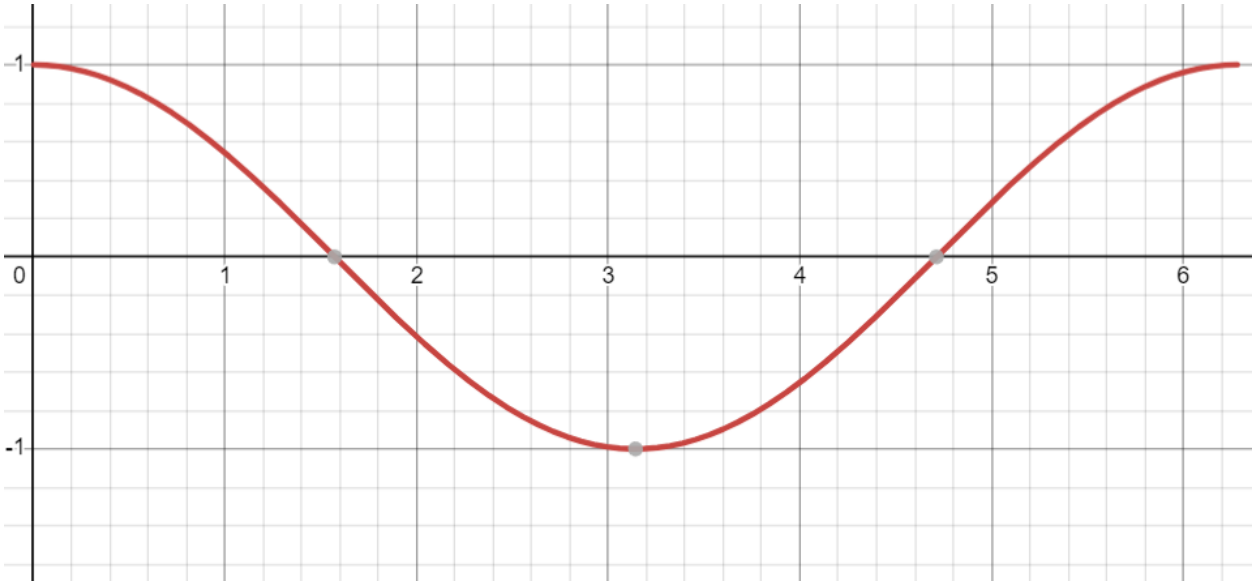
f(x)= sin(x)

3. The function has a flat region near the root: In this case, the derivative of the function is very small near the root, which causes the Newton-Raphson method to converge very slowly or not at all. For example, the function f(x) = x^3 has a flat region near the root at x = 0, which makes it difficult for the Newton-Raphson method to converge.



$f(x) = x^3 \; at \; x = 0$

4. The function has a periodic behavior: In this case, the function repeats itself over a certain interval, which causes the Newton-Raphson method to converge to different roots depending on the initial guess. For example, the function f(x) = cos(x) has a periodic behavior with a period of 2π, which means that the Newton-Raphson method may converge to any one of the infinitely many roots in the interval [0, 2π).



f(x) = cos(x)

b) Draw graphically and explain in words one case where the Secant Root Finding Method will fail. (1.2 points)

The Secant method is another numerical root-finding algorithm that uses an iterative process to approximate the roots of a given function. Here is one case where the Secant method may fail:

The function has a horizontal tangent at the root: In this case, the Secant method may converge very slowly or not at all, because the slope of the function is close to zero at the root. For example, the function f(x) = x^3 has a horizontal tangent at x = 0, which makes it difficult for the Secant method to converge.

f(x) = x^3 at x=0

